A Project on wooden pen holder with drilled holes

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A Project Report related to Course "Engineering Workshop Laboratory" submitted to Mallareddy Engineering College for Women, Hyderabad, in partial fulfillment for the award of Degree of

B. Tech

in

Computer Science Engineering-Data Science

By

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(Autonomous Institution-UGC, Govt. of India)
Accredited by NBA and NAAC with 'A^{+}' Grade

National Ranking by NIRF-Rank band (151-300), MHRD, GOVT. of India
Affiliated to JNTUH, Approved by AICTE, ISO 9001:2015 Certified Institution
Maisammaguda, Dhulapally, Secunderabad-500100

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CERTIFICATE

This is to certify that the **Project Report** related to **Course "Engineering Workshop Laboratory"** titled <A Project on wooden pen holder with drilled holes>submitted by

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of B. Tech I Year I Semester graduates to the **Mallareddy Engineering College for Women, Hyderabad**, in partial fulfillment for the award the degree of **Bachelor of Technology in <Computer Science Engineering-Data Science>** is a *bonafide* record of the project work carried out by them under my/our supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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DECLARATION

We declare that this Project Report related to Course "Computer Aided Engineering Graphics" titled <**A Project on wooden pen holder with drilled holes**> submitted in partial fulfillment of the degree of **Bachelor of Technology in <Computer Science Engineering-Data Science>** is a record of original work carried out by me (*or our Team*) under the supervision of Name of The Faculty-1 and Name of the Faculty-2, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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TITLE PAGE

Project Title:

Design and Fabrication of a Wooden Pen Holder with Drilled Holes

Submitted by:

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Course:

[EWS / Workshop / Engineering Practice]

Institution:

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Instructor:

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Date:

[06/05/2026]

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📘 ABSTRACT

This project focuses on the design and fabrication of a simple and eco-friendly wooden pen holder using drilling techniques. The pen holder is created from a solid wooden block with multiple drilled holes to hold pens, pencils, and other stationery items. The aim of this project is to understand basic woodworking processes such as measuring, cutting, drilling, and finishing. The final product is durable, reusable, and environmentally friendly, promoting sustainable use of materials.

📖 INTRODUCTION

A pen holder is a common desk accessory used to organize writing tools. In this project, a wooden pen holder is designed using basic workshop tools. Wood is chosen as the material due to its availability, ease of machining, and aesthetic appeal.

This project helps in learning:

1.Basic woodworking skills

2.Proper use of drilling machines

3.Measurement and marking techniques

4.Finishing processes

It also encourages sustainable practices by using natural materials instead of plastic.

🧰 MATERIALS AND TOOLS USED

- 🔹 Materials:

1.Wooden block (softwood or hardwood)

2.Sandpaper (coarse and fine grit)

3.Wood polish or varnish

4,Pencil for marking

- Tools:

1.Measuring scale

2.Try square

3.Drill machine

4.Drill bits (various diameters)

5.Clamp or vice

6.Hammer (optional)

7.Brush (for polish)

⚙️ METHODOLOGY

Step 1: Selection of Wood

A suitable wooden block of required dimensions (e.g., 10 cm × 5 cm × 5 cm) is selected.

Step 2: Marking

Use a scale and pencil to mark positions for holes

Ensure equal spacing between holes

Step 3: Clamping

Fix the wooden block securely using a clamp or vice

Step 4: Drilling

Select appropriate drill bit size

Drill vertical holes carefully at marked points

Maintain uniform depth

Step 5: Finishing

Smooth the surface using sandpaper

Remove rough edges

Apply polish or varnish for better appearance

Step 6: Drying

Allow the polish to dry completely

RESULT

The wooden pen holder was successfully fabricated with evenly spaced drilled holes. The final product is:

1.Strong and durable

2.Smoothly finished

3.Aesthetically pleasing

4.Capable of holding multiple pens and pencils

CONCLUSION

The project successfully demonstrated the process of creating a wooden pen holder using basic tools and techniques. It helped in understanding drilling operations and finishing methods. The final product is useful, eco-friendly, and cost-effective. This project enhances practical skills and promotes sustainable material usage.

REFERENCES

1.Workshop Technology by B.S. Raghuwanshi

2.Engineering Workshop Practice Manuals

3.Online resources on woodworking techniques

4.YouTube tutorials on drilling and finishing wood